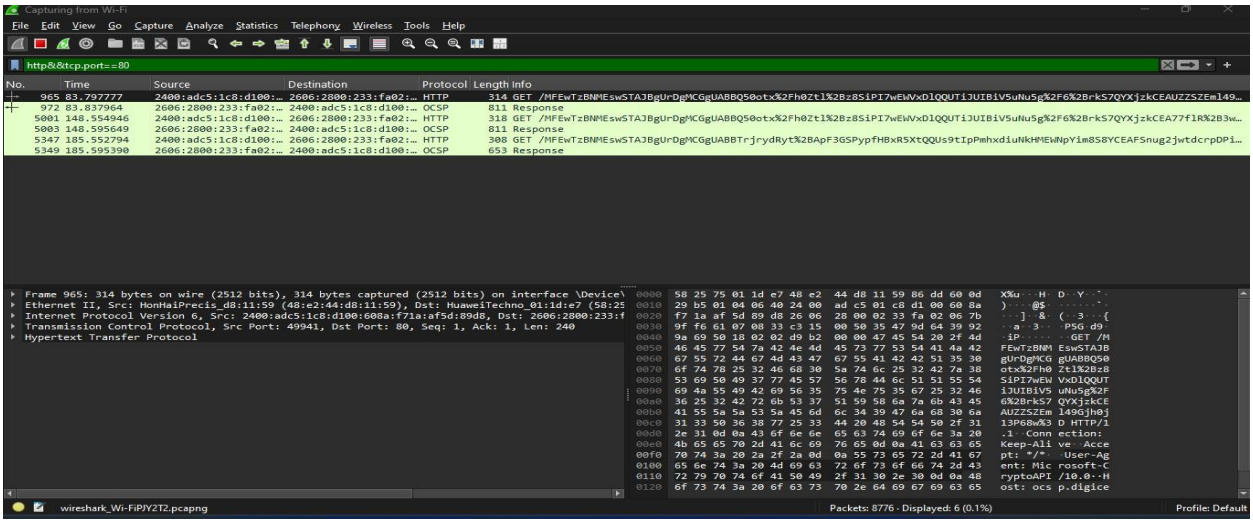
Computer Networks Lab 3

Muhammad Hassan Khalid
21L-5692

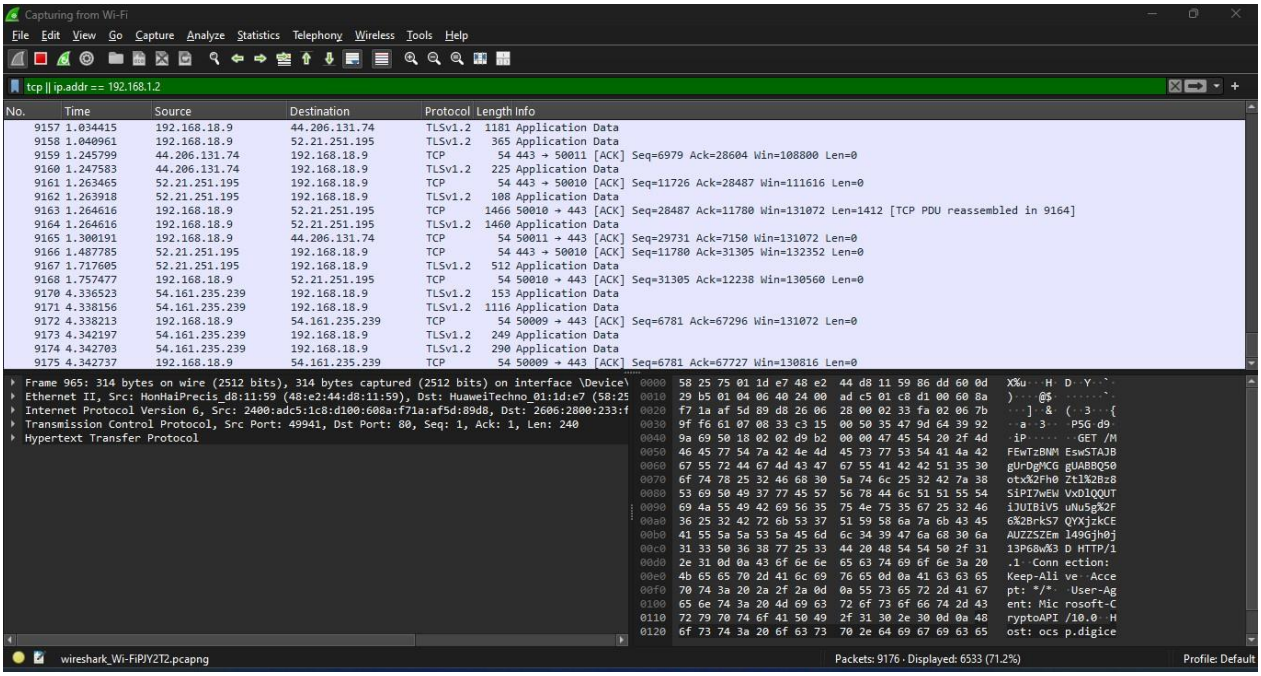
Taking Wireshark for a Test Run

Wireshark Filters

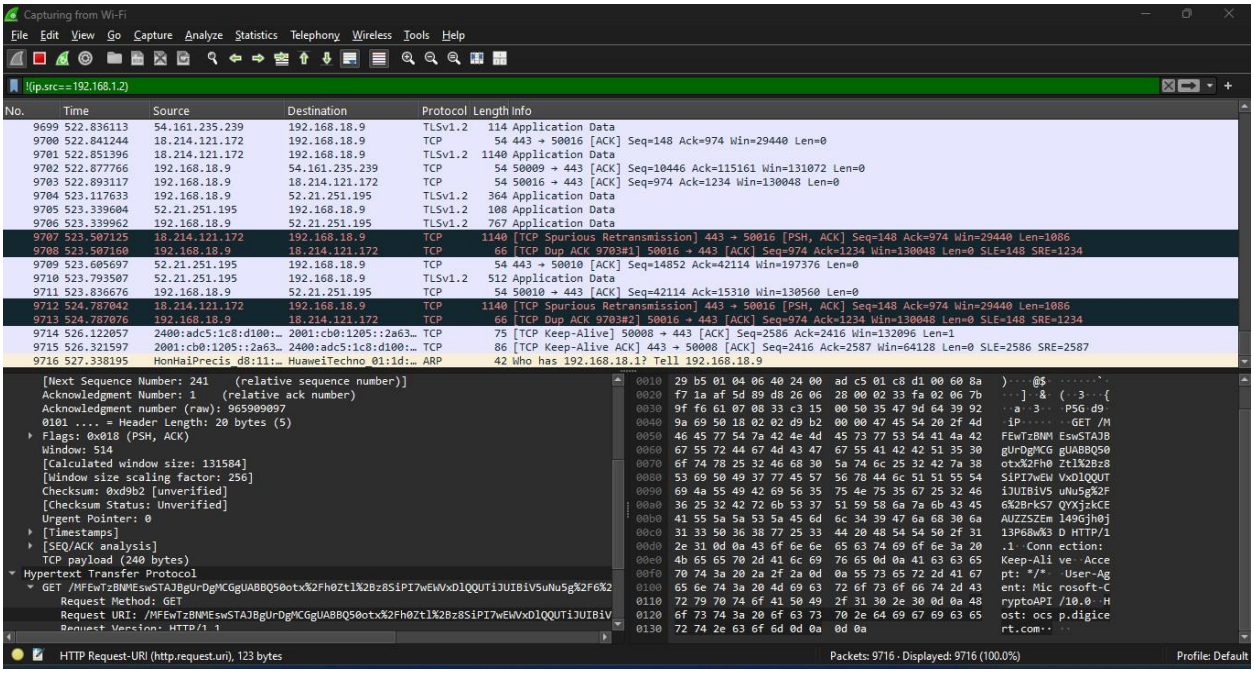
- http && tcp.port==80



- tcp || ip.addr=192.168.1.2



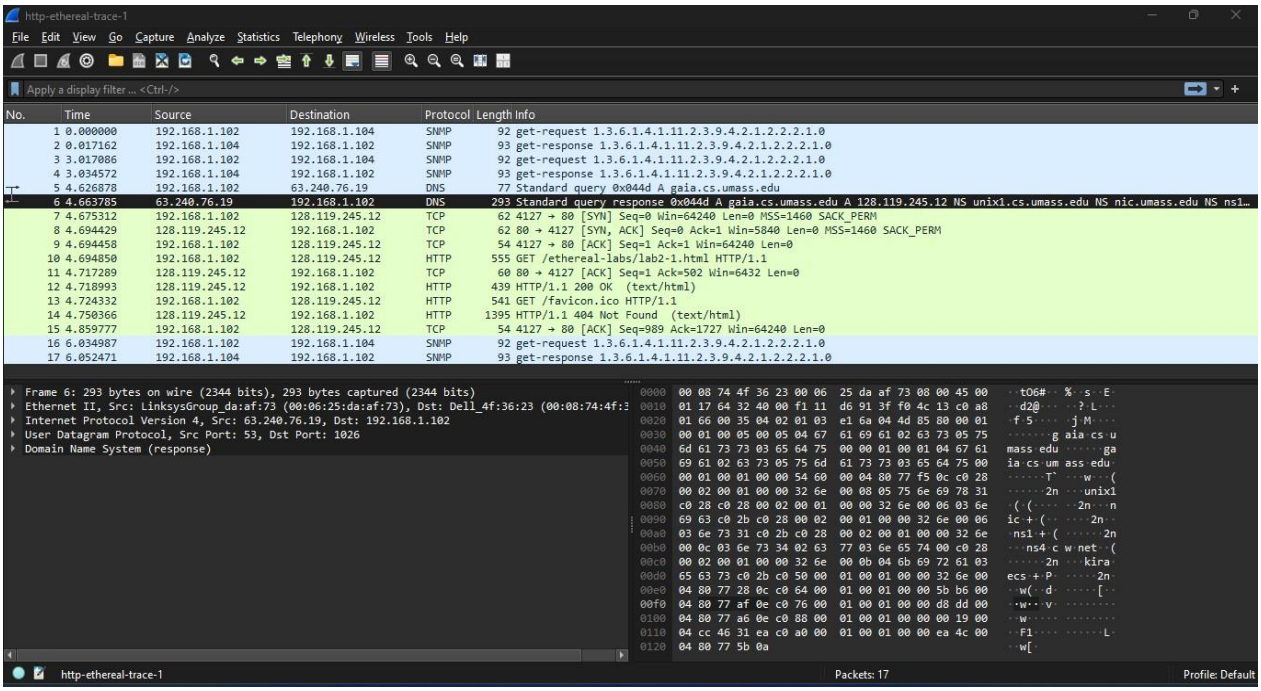
- !(ip.src==192.168.1.2)



In-Lab Statement 1: Analyzing HTTP Protocol

1- **Protocols appearing in the unfiltered packet-listing window:**

- SNMP: Simple Network Management Protocol
- DNS: Domain Name System
- TCP (Transmission Control Protocol)
- HTTP (Hypertext Transfer Protocol)

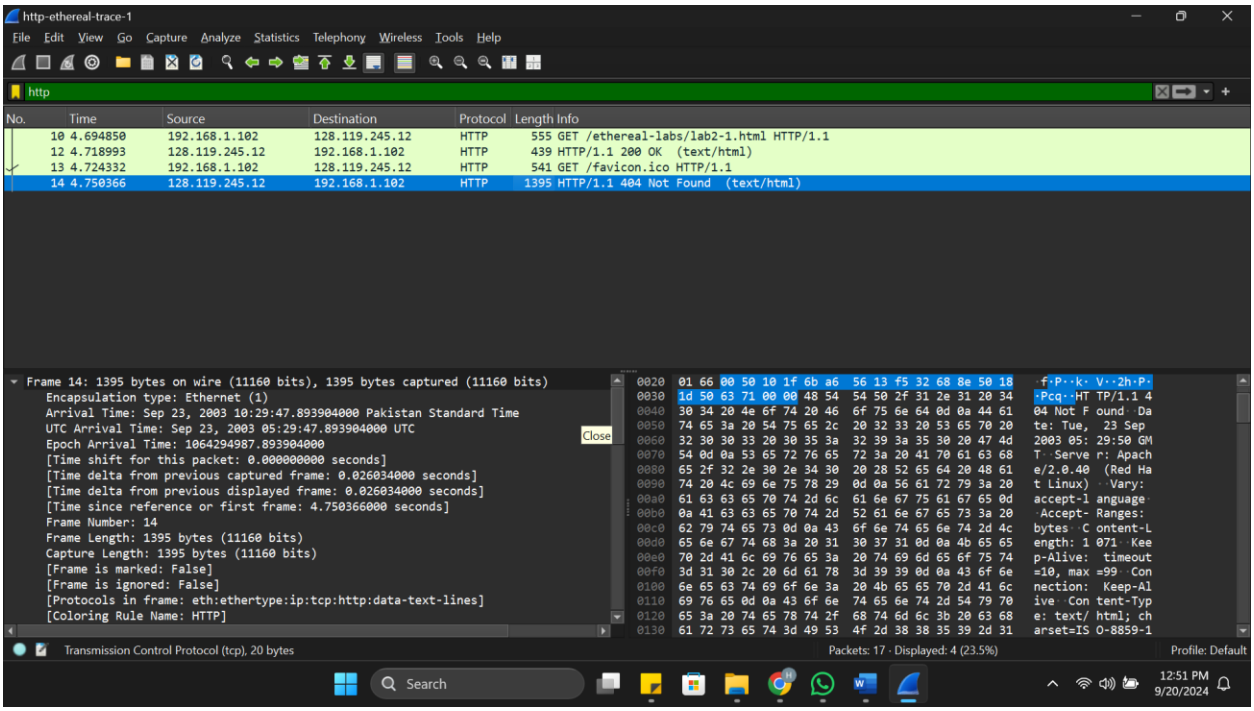


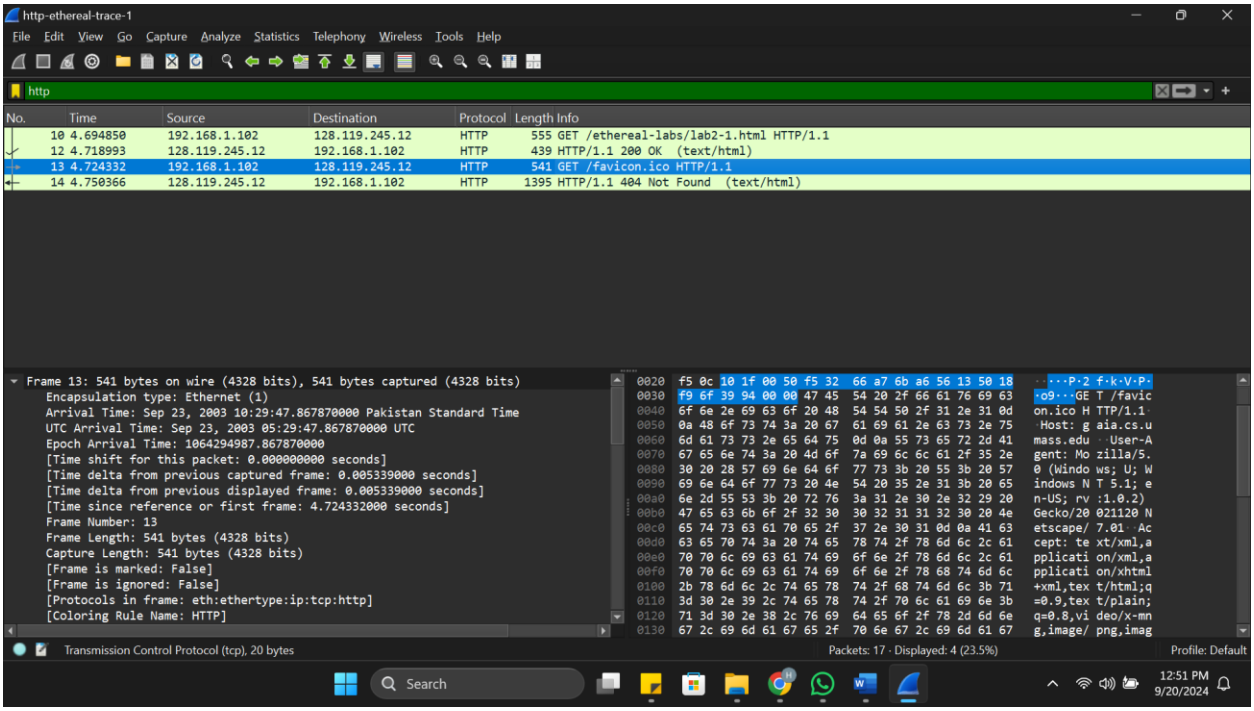
2- **Time taken from HTTP GET to HTTP OK reply:**

From the packet trace provided, the GET request (Frame 13) was sent at 4.724332000 seconds, and the HTTP response (Frame 14) was received at 4.750366000 seconds. Therefore, the round-trip time is approximately:

4.750366−4.724332=0.026034 seconds

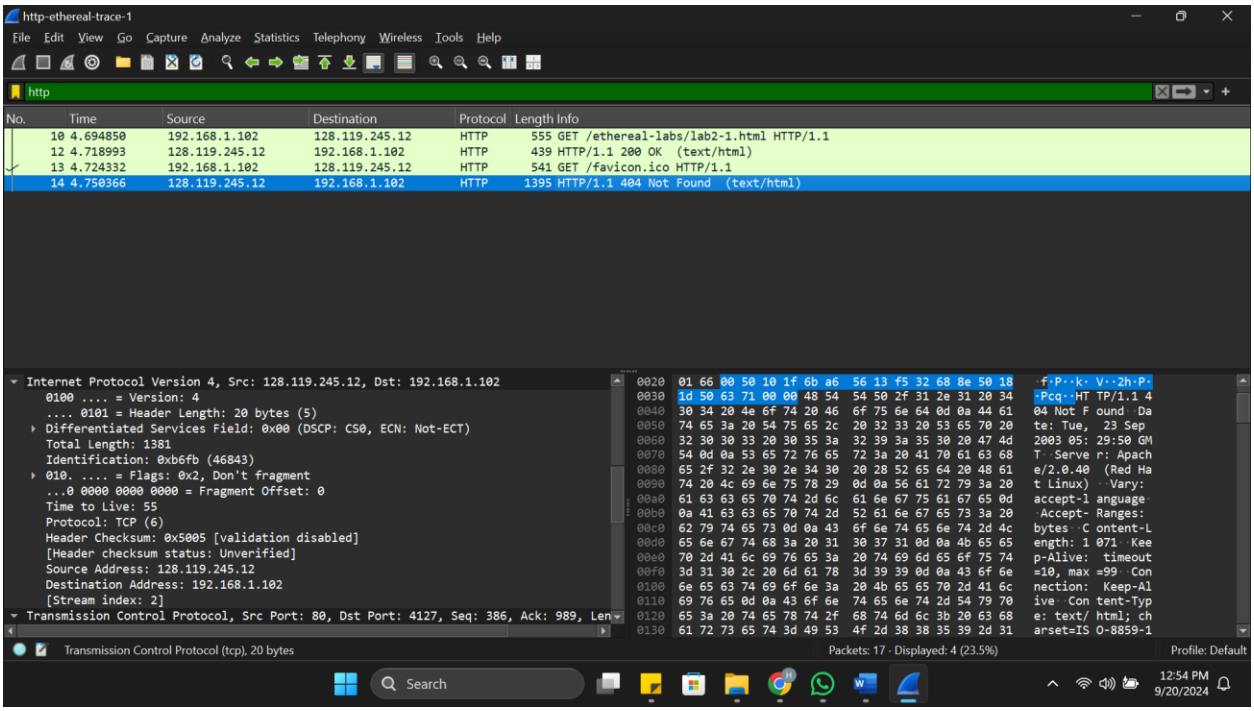
0.026034 * 1000 26.034 ms





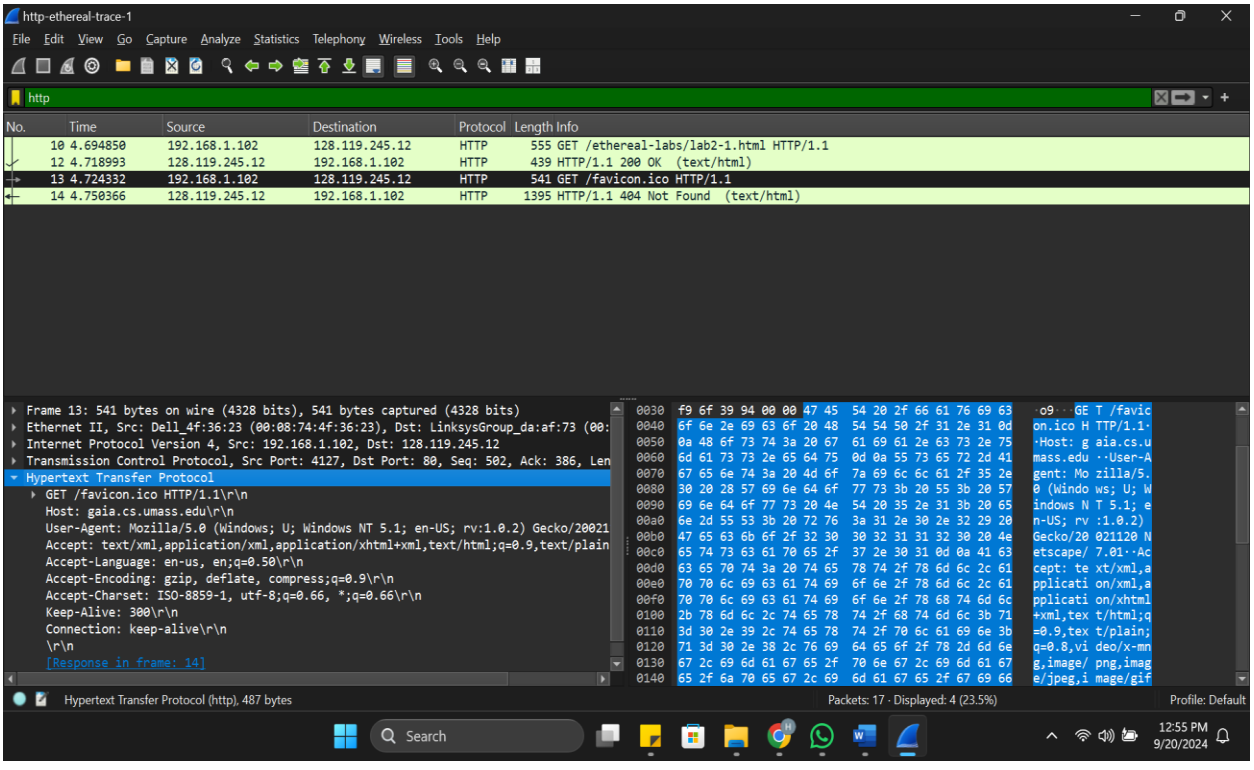
3- **Was the second GET request successful?**

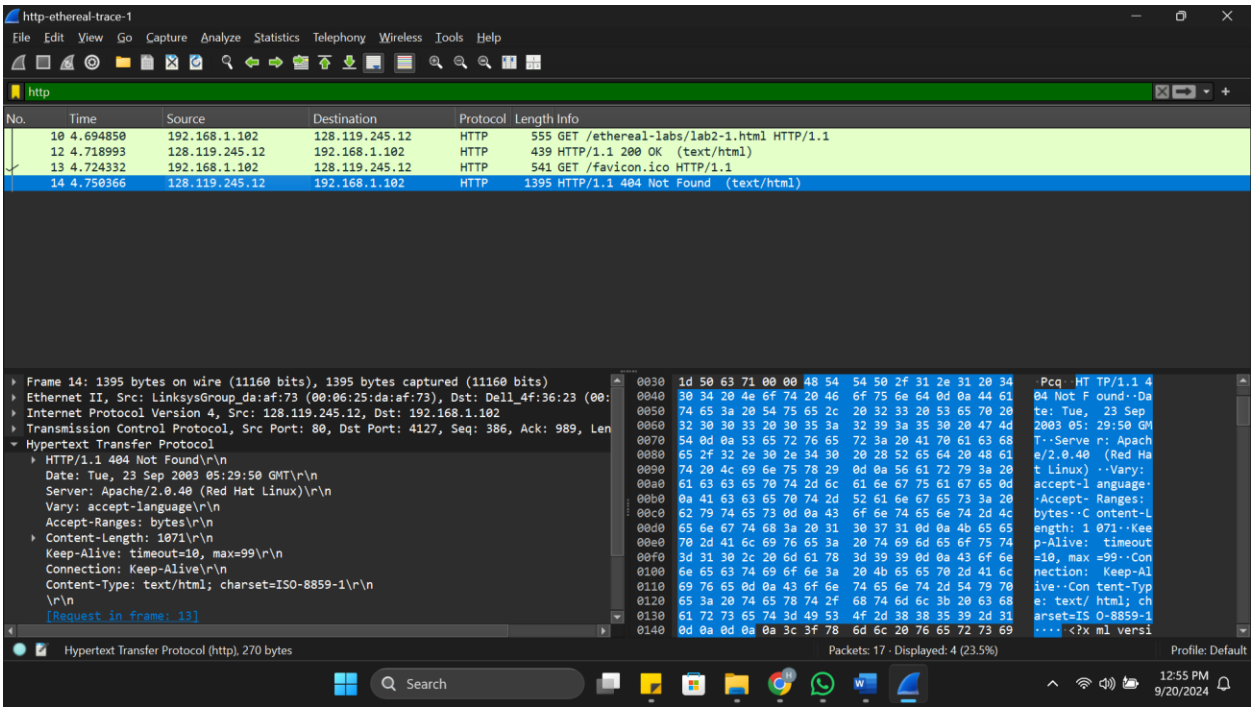
No, the second GET request was not successful. The response to the GET request for /favicon.ico resulted in an HTTP 404 "Not Found" error, as indicated in Frame 14.



4- **HTTP version used by browser and server:**

- **Browser:** The request in Frame 13 shows HTTP/1.1.
- **Server:** The response in Frame 14 also shows HTTP/1.1.



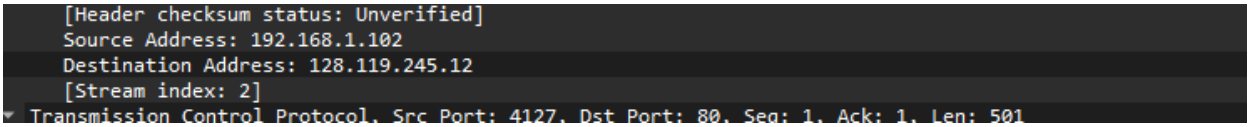


5- **Languages the browser can accept:** From the GET request in Frame 13:

`Accept-Language: en-us, en;q=0.50`

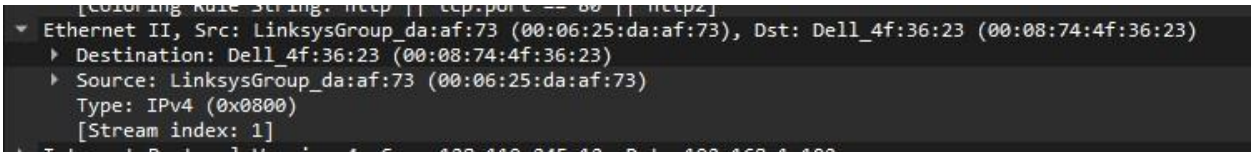
6- **IP addresses:**

- **Computer's IP:** 192.168.1.102
- **gaia.cs.umass.edu server's IP:** 128.119.245.12



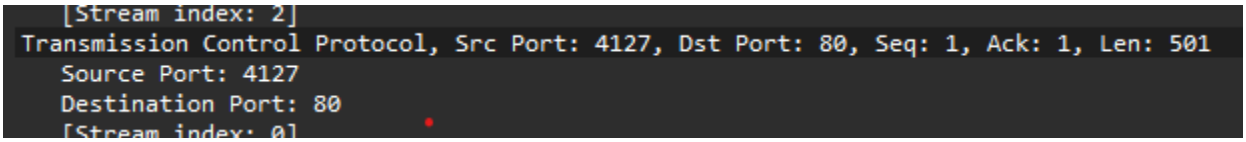
7- **MAC addresses:**

- **Computer's MAC:** `00:08:74:4f:36:23` (Dell_4f:36:23)
- **Server's MAC:** `00:06:25:da:af:73` (LinksysGroup_da:af:73)



8- **Sending and receiving port numbers:**

- **Source Port:** `4127` (Computer)
- **Destination Port:** `80` (the HTTP server)
- Port 80 represents the default port used for HTTP web traffic.



9- **Status code returned from the server:**

The server returned an HTTP 404 status code, meaning the requested resource /favicon.ico was not found.

10- **Last modified date of the HTML file:**

The response does not indicate when the HTML file was last modified, as it is not part of the HTTP 404 error response.

11- **Total bytes returned to the browser:**

The content length in the response header is 1071 bytes, which refers to the body of the response (the HTML error page).

In-Lab Statement 2 : Analyzing HTTP Protocol

The HTTP CONDITIONAL GET/response interaction

1. **Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE” line in the HTTP GET?**

   In the first HTTP GET request (Frame 8), there is **no "IF-MODIFIED-SINCE"** line. This is a standard GET request from the browser to fetch the resource /ethereal-labs/lab2-2.html from the server.

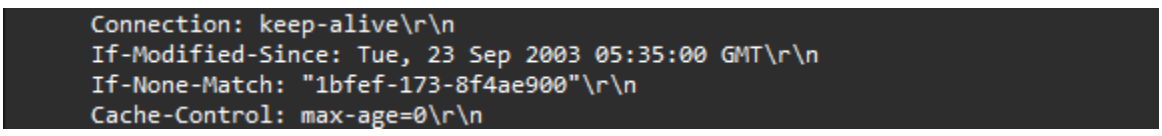
2. **Inspect the contents of the server response. Did the server explicitly return the contents of the file? How can you tell from the Packet Bytes Window?**

   Yes, the server did **explicitly return the contents of the file**. We can tell because the server response (Frame 10) has a status code of **200 OK**, indicating that the requested resource was successfully delivered. Additionally, in the Packet Bytes window, we can see the file contents, including HTML tags like <html> and the message "Congratulations again! Now you've downloaded the file lab2-2.html", confirming the content was transferred.

3. **Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE:” line in the HTTP GET? If so, what information follows the “IF-MODIFIED-SINCE:” header? What is meant by this information?**

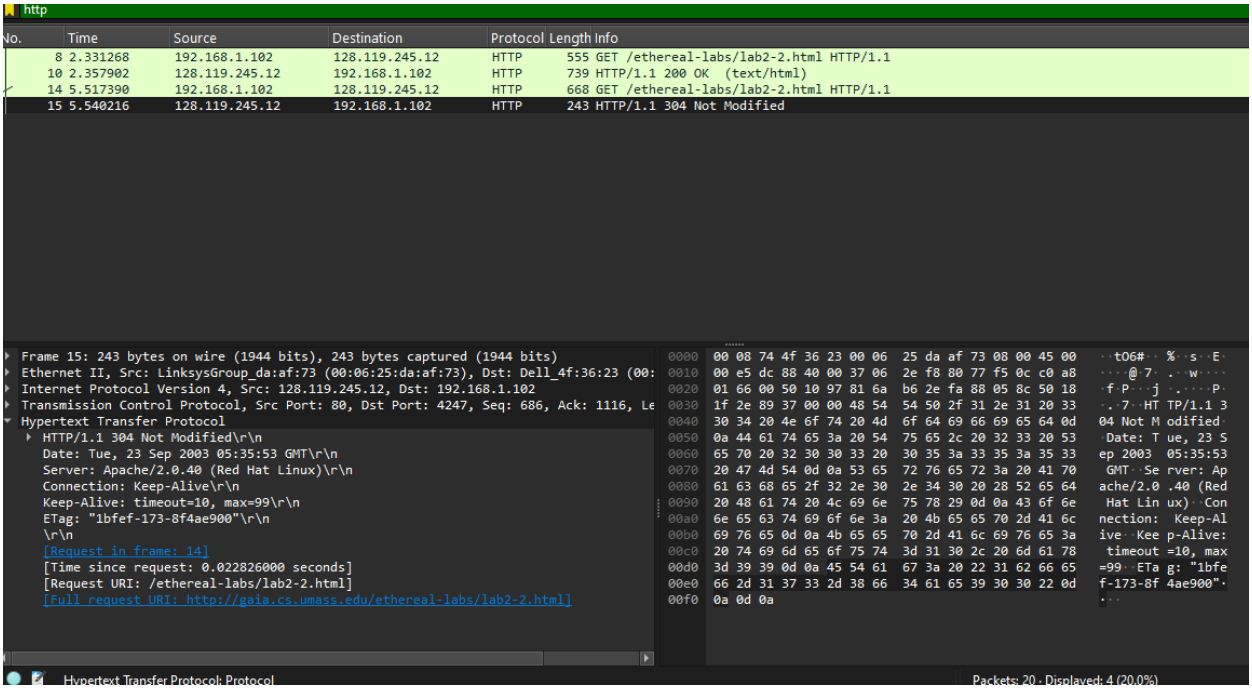
   In the second HTTP GET request (Frame 14), **yes**, there is an "IF-MODIFIED-SINCE" line. The information that follows the header is the timestamp: **Tue, 23 Sep 2003 05:35:00 GMT**.

   This information indicates that the browser is telling the server to send the file only if it has been modified since this timestamp. This is part of the **Conditional GET** mechanism used to optimize resource retrieval by avoiding unnecessary data transfer if the file has not changed.



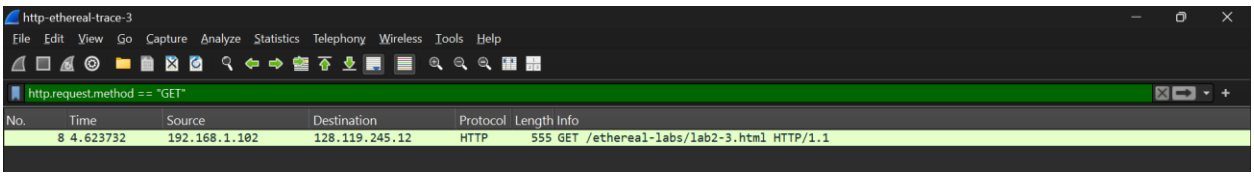
4. **What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain your answer.**

   The server responded with the HTTP status code **304 Not Modified**. This means that the server did **not return the contents of the file** because it determined that the file had not been modified since the specified date and time in the "IF-MODIFIED-SINCE" header. This is why the content is not explicitly returned—only the headers are sent to inform the browser that the cached version can be used.



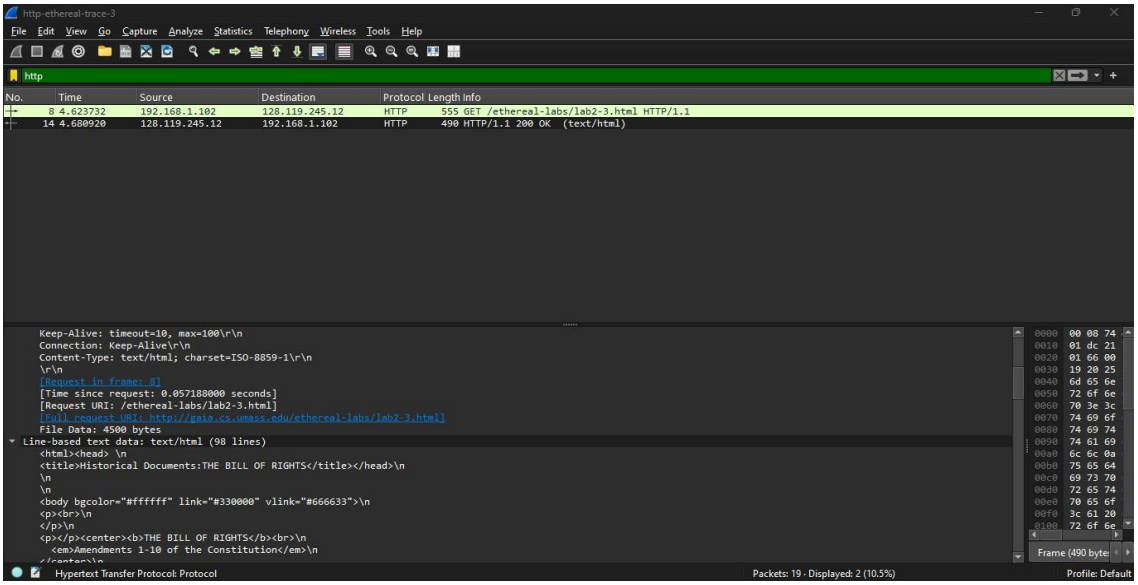
5. **How many HTTP GET request messages did your browser send?**

   To find the number of HTTP GET request messages, you need to filter for http.request.method == "GET". The filtered trace will show that there is **1 HTTP GET request** message sent by the browser.



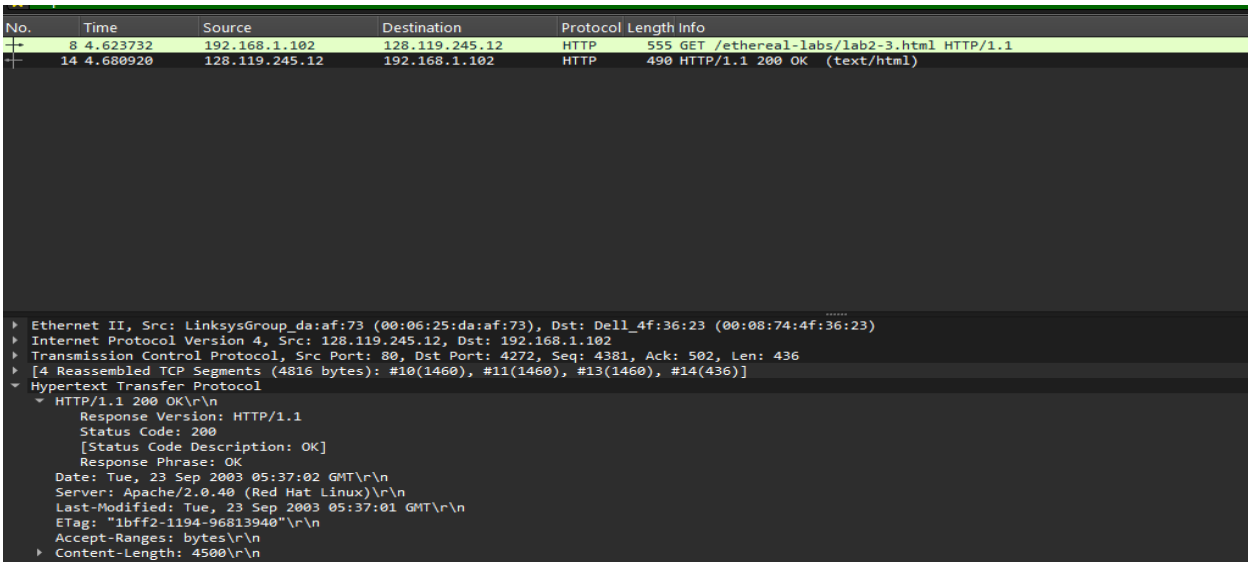
6. Which packet number in the trace contains the GET message for The Bill of Rights?

From the trace details, the **GET message for The Bill of Rights** is contained in **Packet 8**.



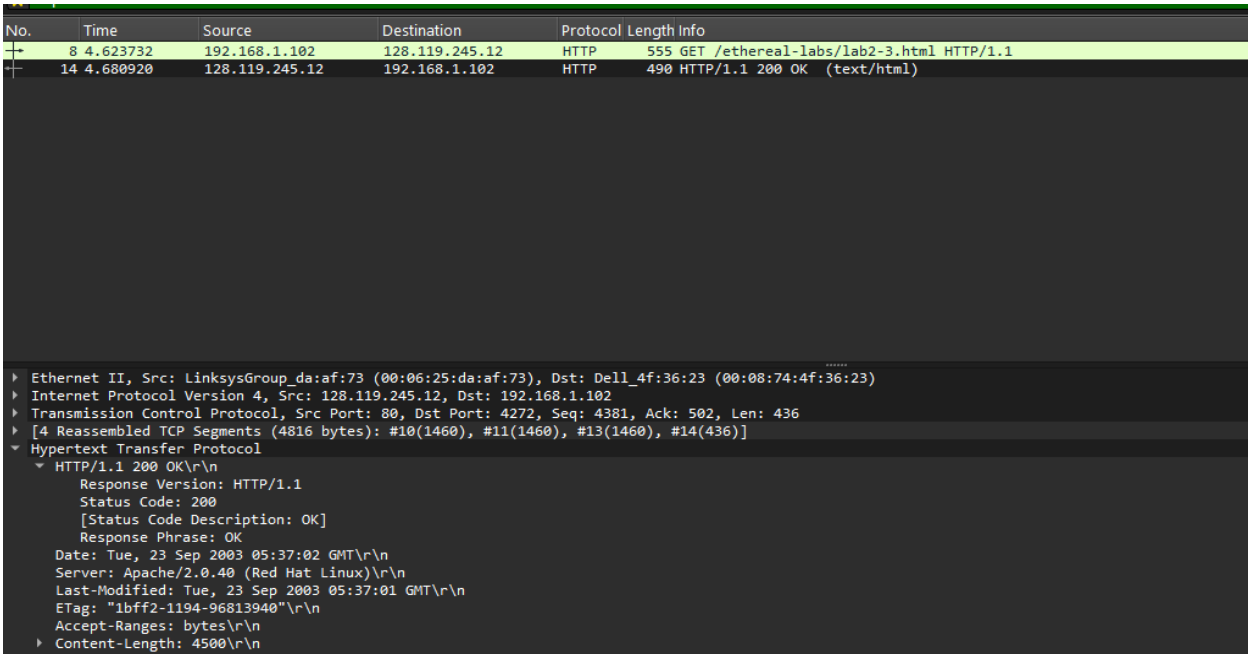
7. Which packet number in the trace contains the status code and phrase associated with the response to the HTTP GET request?

The **status code and phrase in the response** to the HTTP GET request are in **Packet 14**.



8. What is the status code and phrase in the response?

The **status code** in the response is **200**, and the **phrase** is **OK**. This means the request was successful.



9. How many data-containing TCP segments were needed to carry the single HTTP response and the text of the Bill of Rights? What are the numbers of those packets?

The **single HTTP response and the text of the Bill of Rights** are carried in **4 data-containing TCP segments**. These segments are in **Packets 10, 11, 13, and 14**.

In-Lab Statement 3: Trick Question

What is the length of the text for The Bill of Rights in bytes? How do you justify this length of text when your Response Packet Size is only 490 bytes?

- The **length of the text** for The Bill of Rights is **4500 bytes**, as specified in the **Content-Length** header of the HTTP response. However, the response packet in **Packet 14** only contains **490 bytes**.

Explanation:

- The HTTP response is divided across multiple TCP segments. In this case, **Packets 10, 11, 13, and 14** combine to form the complete response, reassembling to a total of **4816 bytes**. Out of these, 4500 bytes represent the actual text of The Bill of Rights, while the remaining bytes account for the HTTP headers.
- The reason the **response packet size is smaller** (490 bytes) is that HTTP data can be broken into smaller segments and transmitted in parts. TCP takes care of assembling these segments upon arrival to deliver the full 4500 bytes of content.